ITCC508 Lab Exercise PT-F1 (Optimizing Chunk Size and Overlap): Initial Preps - Code Review, Test Questions, and Prompt Settings

Section: 401 I

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Objective: To conduct a controlled experiment to determine which chunk size and overlap setting produces the best retrieval performance for a given document dataset using the provided FAISS Vector Store and Retrieval notebook.

Learning Outcomes:

- Create and compare multiple chunk-size and overlap configurations.
- Evaluate retrieval with Precision@3, Recall@3, and MRR@3.
- Compare results for easy, average, and difficult questions.
- Recommend a dataset-specific configuration using measured evidence and efficiency.

Step 1: Code Commenting

Split the code in each notebook cell into individual cells based on logical functional sections. For every code cell, insert a Markdown Text Cell directly above it. Use the Text Cells to provide a step-by-step technical explanation detailing the specific function of each section.

Make sure to preserve the notebook's original functionality and verify that it executes correctly before making moving on with the next steps.

Step 2: Select your dataset

Choose a pdf document appropriate for your group's RAG application. Before proceeding, ensure that:

1. The document contain extractable text.

2. Run the document through the Vector Indexing and Embedding extraction and verify that you are able to generate a valid FAISS vector store by inspecting the chunks in the generated xlsx file.

Use the same dataset throughout the experiment.

Step 2: Prepare the test questions

Create at least three questions based on your own dataset and determine the correct answer and supporting evidence before running the experiments. The Table below will guide you in generating appropriate questions.

Level	Minimum requirement	Example (adapt to your dataset)
Easy	Single explicit fact in one passage	Who is the dean of a specified college?
Average	Interpretation or closely related facts	What requirements apply to a given process?
Difficult	Multiple evidence passages, cross-section reasoning, or entity disambiguation	Compare two policies and identify the conditions under which each applies.

For each question, record:

- The question and its difficulty.
- The expected answer.
- The source document and page, section, or passage containing the evidence.
- All relevant passages if the answer requires multiple pieces of evidence.

Important: All questions must be answerable from your dataset. Difficulty should come from the information needed to answer it, not from vague wording or missing evidence.

Dataset and Recorded Test Questions

Source document: LendSuri_Member_3_Corpus.pdf — the Member 3 corpus of the LendSuri PT-F1 single-PDF package (Group QuantumStack). 126 pages, one researcher-prepared record per page, summarising Philippine SEC records on lending companies and online lending platforms: the January 26, 2026 registry snapshot (pages 1–60), the November 29, 2021 historical list (pages 61–101), SEC regulatory events and advisories (pages 102–123), framework guidance (pages 124–125) and complaint guidance (page 126).

Preprocessing: every page has a text layer, so pypdf extracts all 126 pages and no OCR was needed. Each page, however, repeats the same running header ("LendSuri | Researcher-prepared record from the existing v5 corpus"), a printed page number that belongs to the original category PDF rather than to this file, the label "LendSuri / Member 3", and a two-line disclaimer footer — 34,368 of 164,290 characters, 20.9% of the corpus. These were removed, because at small chunk sizes they would fill much of each chunk and bias the experiment toward large chunks, and the printed page number would mislead source references. The cleaned text was re-rendered one record per page as LendSuri_Member_3_Corpus_clean.pdf (129,983 characters after whitespace normalisation), so every record keeps its page number and the package's gold pages apply unchanged. The answer key, question sheet and page map were never uploaded or indexed. Verified in Component I: 126 pages extracted, and 235 chunks in a 235-vector FAISS index at the 1000 / 200 baseline, inspected in the exported chunk workbook.

Question 1 — Easy (M3-E)

Question: Which lending company is Peso2you listed under in the SEC records?

Expected answer: The January 26, 2026 snapshot associates Peso2you with Snapcash Lending, Inc.

Evidence: page 46 (record CUR2026_0141) — single passage. "Platform: Peso2you. Company: Snapcash Lending, Inc."

Question 2 — Average (M3-A)

Question: BillEase appears under First Digital Finance in SEC records, but I saw another lender using that company name. Has the SEC warned about this?

Expected answer: The January 2026 snapshot lists BillEase under First Digital Finance Corporation. A May 21, 2026 advisory entry concerns an unregistered lender or unrecorded OLP using the company's identity. The advisory must not automatically be attributed to BillEase itself.

Evidence: two passages on two pages — page 15 (CUR2026_0053): "Platform: Billease. Company: First Digital Finance Corporation."; page 109 (EVT0030): "Public advisory on unauthorized use of First Digital Finance Corporation by an unregistered lending company and unrecorded OLP."

Question 3 — Difficult (M3-D)

Question: I'm considering Tala, but I've also seen a Tala Facebook page and another Tala-branded loan app. Is Tala in SEC records, and what warnings should I know about?

Expected answer: The January 2026 registry snapshot includes Tala under Tala Financing Philippines Inc. A February 12, 2026 advisory entry concerns an unauthorized Facebook page impersonating Tala, and a June 9, 2026 advisory entry concerns fraudulent OLPs/mobile apps misusing its name. The dated listing does not authenticate a particular page or app.

Evidence: three passages across two record types and three pages — page 54 (CUR2026_0156, January 2026 registry), page 105 (EVT0017, February 12, 2026 advisory on an unauthorized Facebook page impersonating Tala) and page 111 (EVT0033, June 9, 2026 advisory on fraudulent OLPs and mobile apps using Tala's name). Requires combining a registry entry with two later advisories and keeping the registered company separate from its impersonators.

Note on distractors: the 2021 historical list also names these platforms under the same companies — Peso2you on page 95, BillEase on page 72 and Tala on page 97. These pages are on-topic but are not the evidence the answer key requires, so they are not counted as relevant.

3. Testing the Language Model

The existing notebook uses google/flan-t5-base with a baseline chunk size of 1,000 characters and a 200-character overlap.

Your Task:

1. Update the code to use your group’s assigned language model.
2. Adjust the prompt and implement necessary guardrails to ensure the model generates accurate answers.

Note: If your assigned model is unable to produce correct answers, you are free to evaluate alternative models.

4. Submit results

Indicate below the final language model and prompt that worked on your dataset and submit this file along with your commented notebook:

- **Final Language Model:** **aisingapore/Llama-SEA-LION-v3-8B-IT**

- **Final Prompt:** (assembled with the retrieved evidence by Component II section 6 and wrapped in the model's chat template)

  Answer the question using only the supplied evidence records. Every record is dated: give the date and the record type (2026 registry snapshot, 2021 historical list, SEC advisory, framework guidance or complaint guidance) for each fact you use. Write platform and company names exactly as they appear. A registry listing only shows that a platform was recorded on that date; it does not prove current authorization and does not show that any Facebook page, app or website is genuine. An advisory about unauthorized use of a company's name concerns the impersonator, not the registered company, so do not attribute it to the company. Answer each part of a multi-part question separately and name any part the evidence does not cover. Do not guess or use outside knowledge. If the evidence does not establish the answer, respond exactly: 'I cannot find the answer in the provided document.'

Experiment Results

Five configurations were tested: chunk sizes 300 / 1000 / 1800 at overlap 200, then overlaps 0 / 200 / 450 with chunk size held at 1000. Top-K (3), the embedding model (all-MiniLM-L6-v2), the three Member 3 questions and the cleaned dataset were held fixed throughout. A retrieved chunk counted as relevant only when it contained a required evidence passage in full, located by its exact text on the gold page, so the same evidence is tracked even though chunk numbering changes with every setting. Scores were computed with the same chunking and scoring functions as Component III of the PT-F2 notebook.

Configuration	Chunks	Precision@3	Recall@3	MRR@3
300 / 200	1370	0.222	0.278	0.278
1000 / 0	207	0.444	0.500	0.667
1000 / 200 (baseline)	235	0.444	0.500	0.667
1800 / 200 (selected)	**126**	**0.444**	**0.500**	**0.667**
1000 / 450	320	0.333	0.389	0.667

Recommended configuration: chunk size 1800, overlap 200.

1000 / 0, 1000 / 200 and 1800 / 200 tie on all three metrics (Precision@3 0.444, Recall@3 0.500, MRR@3 0.667), so the choice rests on efficiency: 1800 / 200 needs 126 chunks — one per record page — against 207 and 235. 300 / 200 is clearly worst (1,370 chunks, Recall@3 0.278): its chunks split records so the platform and company often land in different chunks. Raising overlap to 450 added 85 chunks over the baseline and lowered precision, because overlapping chunks of the same page crowded the top three. With only three questions the tie is not a demonstrated optimum; Part 2 tests nine questions and twelve configurations.

Answer accuracy at the baseline configuration (1000 / 200)

Run in Colab on a T4 GPU with the model loaded in 4-bit NF4. All three retrieved chunks fitted the prompt on every question (786, 631 and 963 tokens against a 4,096-token budget).

Question 1 (M3-E): No answer — a correct refusal. Retrieval returned pages 99, 121 and 32, none of which holds the Peso2you record (page 46), and the model replied "I cannot find the answer in the provided document." instead of guessing a company. Page 46 was not retrieved at any of the five tested configurations (Recall@3 = 0 for this question throughout).

Question 2 (M3-A): Partially correct. It reported the May 21, 2026 advisory on an unregistered lending company using the First Digital Finance name and, as the answer key requires, did not attribute the advisory to BillEase. The BillEase listing on page 15 was not retrieved (pages 109, 32 and 59 were), so the model stated that the evidence does not cover BillEase rather than asserting the listing.

Question 3 (M3-D): Correct. All three gold pages were retrieved (111, 105 and 54). The answer gives the January 26, 2026 listing under Tala Financing Philippines Inc., the February 12, 2026 Facebook-impersonation advisory and the June 9, 2026 fraudulent-apps advisory, and adds that the advisories do not establish whether any particular page or app is genuine.

On guardrails and their limit

The prompt adds four dataset-specific rules to the grounding guardrails: date and type every record used; treat a registry listing as evidence of recording only, not of current authorization or of a page's authenticity; never attribute an impersonation advisory to the registered company; and answer each part separately while naming any part the evidence does not cover. Each rule showed up in the answers — the refusal on M3-E, the non-attribution and gap-flagging on M3-A, and the authenticity caveat on M3-D. The guardrails cannot recover evidence that retrieval does not supply: on two of the three questions the model was limited by missing gold pages, not by its reasoning.

Dataset-specific finding

Every record page shares the same template, so a chunk's embedding is dominated by wording common to all 126 records, and only the platform name, company and dates distinguish one record from another. Queries that name a product therefore retrieve structurally similar records for other products: at the baseline the Peso2you query ranked the PesoQ record (page 99) first. The difficult Tala question, whose wording ("Facebook page", "app", "warnings") matches the advisory records' own vocabulary, retrieved all three of its gold pages.